Open Source Audit – Python

STUDENT NAME	SARANG KULKARNI
REGISTRATION NUMBER	24BOE10014
SLOT	E22
DATE OF SUBMISSION	31-03-2026

1. Introduction

Open-source means that the source code of a software is available for anyone to look at, use, modify and share. This is different from proprietary software, where the code is private and controlled by a company. Open-source software is made in a way that is transparent and collaborative. Developers from all over the world can work together to improve the software, fix problems and add new features. This makes open source not just a type of software but also a way of thinking about sharing knowledge and working together.

Open source is very important today because it is used in almost everything we do with technology. From operating systems like Linux to programming languages like Python, open-source tools form the base of software development. It also makes technology more accessible because people do not have to pay a lot of money to use these tools. In my view, open source also helps people learn because you can see how programs are made instead of just using them without knowing how they work.

This project is about learning more about open source by studying Python. Instead of just learning what Python can do, the goal is to find out why it was created, what problems it solves, what its license says and how it works with Linux. This project also includes writing shell scripts to show that I understand how open-source tools are used in systems. The goal of this project is not only to learn about Python but also to understand the bigger idea behind open-source software and why it is so important in the technology world.

---

2. The Audit

Part A — Origin and Philosophy

A1. The problem this software was created to solve

Before Python came into the picture, programming was not very beginner-friendly. Programming languages like C and C++ were powerful, but they were also quite complicated. Writing programs required a lot of syntax and attention to detail, which could be frustrating, especially for beginners. It felt like you had to focus more on how to write the code rather than what you actually wanted the program to do.

Python was created by Guido van Rossum in 1991 in the Netherlands. His main idea was simple: he wanted to create a programming language that was easy to read and easy to use. Instead of making things more complex, he focused on simplicity and clarity. Python uses clean and readable syntax. Unlike other languages that use many symbols and complicated structures, Python uses indentation and simple words, which makes code look almost like normal English.

Another important thing about Python is that it was released as an open-source project. This means anyone could access it, modify it and contribute to it. Because of this, a large community of developers helped improve Python over time. Today, Python is used in many fields such as web development, artificial intelligence, data science, and automation.

Overall, Python started as an attempt to simplify programming, and over time it has become one of the most important programming languages in the world.

A2. The license — what it actually says

Python is released under the Python Software Foundation License. This is an open-source license that allows people to use, modify and distribute the software freely. The license is very flexible and does not have many strict rules. People can use Python for personal projects, education, and even commercial applications.

The four main freedoms of open-source software are:

1. Freedom to use the software for any purpose

2. Freedom to study how the software works

3. Freedom to modify the software

4. Freedom to distribute the original or modified software

Python's license allows all these freedoms, which makes it a true open-source software.

A3. The ethics of open source

Open source is not just about sharing code but also about sharing knowledge and technology. A common question is whether all software should be open source. On one hand, open source makes software more accessible, transparent and collaborative. On the other hand, proprietary software is also important for businesses.

Companies like Google and Netflix use open-source technologies like Python and Linux. They benefit from open source, but they also contribute back to the community. This creates a balanced relationship.

In my opinion, both open-source and proprietary software are important. Open source helps learning and innovation, while proprietary software helps businesses and provides professional support.

Part B — Linux Footprint

This section shows how Python is structured and managed within a Linux system.

Installation Method

Python is available through the apt package manager in Linux. In most systems, Python is already installed by default. It can also be installed using the command:

sudo apt install python3

This shows that Python is distributed as a package and does not require manual compilation.

```
loq@SARANG:~$ python3 --version
Python 3.12.3
```

Figure1: python3 --version

Key Directories

The Python executable is located at /usr/bin/python3, which can be verified using the command:

which python3

Python does not maintain a central configuration file in /etc and does not usually create logs in /var/log because it is a programming language, not a system service.

```
loq@SARANG:~$ which python3
/usr/bin/python3
```

Figure2: which python3

```
loq@SARANG:~$ dpkg -l | grep '^ii  python3 '
ii  python3                      3.12.3-0ubuntu2.1                      am
d64        interactive high-level object-oriented language (default python3
version)
```

User and Group

Python runs under the current user account. Permissions can be checked using:

ls -l /usr/bin/python3

This is important for system security because running scripts as a normal user prevents damage to system files.

Service Management

Python is not a system service. It runs only when a script is executed. However, applications built using Python (like Django or Flask web apps) can be managed as services using systemctl.

Update Model

Python is updated using the package manager:

sudo apt update

sudo apt upgrade

```
loq@SARANG:~$ dpkg -l | grep '^ii  python3 '
ii  python3                      3.12.3-0ubuntu2.1                      am
d64        interactive high-level object-oriented language (default python3
version)
loq@SARANG:~$ dpkg -l | grep libpython3
ii  libpython3-stdlib:amd64      3.12.3-0ubuntu2.1                      am
d64        interactive high-level object-oriented language (default python3
version)
ii  libpython3.12-minimal:amd64   3.12.3-1ubuntu0.11                    am
d64        Minimal subset of the Python language (version 3.12)
ii  libpython3.12-stdlib:amd64    3.12.3-1ubuntu0.11                    am
d64        Interactive high-level object-oriented language (standard library
, version 3.12)
ii  libpython3.12t64:amd64        3.12.3-1ubuntu0.11                    am
d64        Shared Python runtime library (version 3.12)
```

Part C — The FOSS Ecosystem Dependencies

Python depends on many open-source components such as system libraries, compilers and package repositories. Tools like pip depend on network libraries and online repositories. This shows that open-source projects depend on each other and work together like an ecosystem.

Python is used with many libraries such as NumPy, Pandas, Django, Flask and TensorFlow. This makes Python useful in data science, web development, automation and machine learning.

Part D — Open-Source vs Proprietary (Critical Analysis)

Dimension	Python (Open Source)	MATLAB (Proprietary)
Cost	Free	Paid
Security	Open source code	Closed source
Support	Community support	Official support
Modification	Can modify	Cannot modify
Control	Community	Company

After comparing Python and MATLAB, both have advantages. Python is free and flexible, while MATLAB provides professional tools and support. As a student, Python is more practical and accessible.

---

3. Shell Scripts

Script 1 – System Identity Report

Author: Sarang Kulkarni

```
(script1.sh)
#! /bin/bash

# Script 1: System Identity Report
# Author: Sarang Kulkarni
```

```
# Course: Open-Source Software

# --- Variables ---
STUDENT_NAME="Sarang Kulkarni"

# --- System Info ---
KERNEL=$(uname -r)
DISTRO=$(lsb_release -d | cut -f2)
USER_NAME=$(whoami)
HOME_DIR=$HOME
UPTIME=$(uptime -p)
CURRENT_DATE=$(date)

# --- Display --- echo
"==================================
====" echo "   Open Source Audit -
$STUDENT_NAME" echo
"==================================
===="
 echo "Distribution :
$DISTRO" echo "Kernel
: $KERNEL" echo "User
: $USER_NAME" echo "Home
Dir     : $HOME_DIR" echo
"Uptime       : $UPTIME"
echo "Date/Time    :
$CURRENT_DATE"
 echo "--------------------------------------" echo "This system
is typically licensed under" echo "the GNU General Public License
(GPL)." echo "--------------------------------------"
```

```
loq@SARANG:/mnt/c/Users/LOQ$ cd ~
loq@SARANG:~$ pwd
/home/loq
loq@SARANG:~$ nano script1.sh
loq@SARANG:~$ chmod +x script1.sh
loq@SARANG:~$ ./script1.sh
==================================
Open Source Audit — Sarang Kulkarni
==================================
Software : Python
Kernel   : 6.6.87.2-microsoft-standard-WSL2
User     : loq
Uptime   : up 16 minutes
Date     : Mon Mar 30 18:48:53 UTC 2026
Distro   : "Ubuntu 24.04.4 LTS"
License  : Linux is covered under GPL license
loq@SARANG:~$ |
```

```
loq@SARANG:~$ (script1.sh)
#! /bin/bash

# Script 1: System Identity Report
# Author: Sarang Kulkarni
# Course: Open-Source Software

# --- Variables ---
STUDENT_NAME="Sarang Kulkarni"

# --- System Info ---
KERNEL=$(uname -r)
DISTRO=$(lsb_release -d | cut -f2)
USER_NAME=$(whoami) HOME_DIR=$HOME
UPTIME=$(uptime -p)
CURRENT_DATE=$(date)

# --- Display --- echo "==================================" echo "   Open Source Audit - $STUDENT_NAME" echo "====================
=================="
 echo "Distribution : $DISTRO" echo "Kernel       : $KERNEL" echo "User         : $USER_NAME" echo "Home Dir     : $HOME_DIR" echo "Up
time       : $UPTIME" echo "Date/Time    : $CURRENT_DATE"
 echo "------------------------------------" echo "This system is typically licensed under" echo "the GNU General Public License (GP
L)." echo "------------------------------------"

script1.sh: command not found
Distribution : Ubuntu 24.04.4 LTS echo Kernel       : 6.6.87.2-microsoft-standard-WSL2 echo User         : loq echo Home Dir     : /ho
me/loq echo Uptime       : up 4 hours, 20 minutes echo Date/Time    : Tue Mar 31 17:24:16 UTC 2026
------------------------------------ echo This system is typically licensed under echo the GNU General Public License (GPL). echo --
------------------------------------
```

This script displays system information such as kernel version, username, uptime, date and Linux distribution. It uses Linux commands and variables to display system information.

Script 2 – FOSS Package Inspector

Author: Sarang Kulkarni

```
#!/bin/bash

PACKAGE="python3"


if dpkg -l | grep -q $PACKAGE; then

   echo "$PACKAGE is installed."

   dpkg -l | grep $PACKAGE

else

   echo "$PACKAGE is NOT installed."

fi


case $PACKAGE in

   python3) echo "Python: a powerful open-source programming language used everywhere" ;;

   firefox) echo "Firefox: open web browser" ;;

   vlc) echo "VLC: plays all media formats" ;;
```

```
  git) echo "Git: version control system" ;;

esac
```

```
ii  python3-typing-extensions     4.10.0-1                          all        Backported and Experimental Type Hints for Python
ii  python3-tz                    2024.1-2                          all        Python3 version of the Olson timezone database
ii  python3-update-manager        1:24.04.12                        all        Python 3.x module for update-manager
ii  python3-urllib3               2.0.7-1ubuntu0.6                  all        HTTP library with thread-safe connection pooling for Python3
ii  python3-wadllib               1.3.6-5                           all        Python 3 library for navigating WADL files
ii  python3-yaml                  6.0.1-2build2                     amd64      YAML parser and emitter for Python3
ii  python3-zope.interface        6.1-1build1                       amd64      Interfaces for Python3
ii  python3.12                    3.12.3-1ubuntu0.11                amd64      Interactive high-level object-oriented language (version 3.12)
ii  python3.12-minimal            3.12.3-1ubuntu0.11                amd64      Minimal subset of the Python language (version 3.12)
Python: a powerful open-source programming language used everywhere
log@SARANG:~$
```

```
ii  python3-hyperlink             21.0.0-5                          all        Immutable, Pythonic, correct URLs.
ii  python3-idna                  3.6-2ubuntu0.1                    all        Python IDNA2008 (RFC 5891) handling (Python 3)
ii  python3-incremental           22.10.0-1                         all        Library for versioning Python projects
ii  python3-jinja2                3.1.2-1ubuntu1.3                  all        small but fast and easy to use stand-alone template engine
ii  python3-json-pointer          2.0-0ubuntu1                      all        resolve JSON pointers - Python 3.x
ii  python3-jsonpatch             1.32-3                            all        library to apply JSON patches - Python 3.x
ii  python3-jsonschema            4.10.3-2ubuntu1                   all        An(other) implementation of JSON Schema (Draft 3, 4, 6, 7)
ii  python3-jwt                   2.7.0-1                           all        Python 3 implementation of JSON Web Token
ii  python3-launchpadlib          1.11.0-6                          all        Launchpad web services client library (Python 3)
ii  python3-lazr.restfulclient    0.14.6-1                          all        client for lazr.restful-based web services (Python 3)
ii  python3-lazr.uri              1.0.6-3                           all        library for parsing, manipulating, and generating URIs
ii  python3-markdown-it           3.0.0-2                           all        Python port of markdown-it and some its associated plugins
ii  python3-markupsafe            2.1.5-1build2                     amd64      HTML/XHTML/XML string library
ii  python3-mdurl                 0.1.2-1                           all        Python port of the JavaScript mdurl package
ii  python3-minimal               3.12.3-0ubuntu2.1                 amd64      minimal subset of the Python language (default python3 version)
ii  python3-netifaces:amd64       0.11.0-2build3                    amd64      portable network interface information - Python 3.x
ii  python3-netplan               1.1.2-8ubuntu1~24.04.1            amd64      Declarative network configuration Python bindings
ii  python3-newt:amd64            0.52.24-2ubuntu2                  amd64      NEWT module for Python3
ii  python3-oauthlib              3.2.2-1                           all        generic, spec-compliant implementation of OAuth for Python3
ii  python3-openssl               23.2.0-1                          all        Python 3 wrapper around the OpenSSL library
ii  python3-pkg-resources         68.1.2-2ubuntu1.2                 all        Package Discovery and Resource Access using pkg_resources
ii  python3-problem-report        2.28.1-0ubuntu3.8                 all        Python 3 library to handle problem reports
ii  python3-pyasn1                0.4.8-4ubuntu0.1                  all        ASN.1 library for Python (Python 3 module)
ii  python3-pyasn1-modules        0.2.8-1                           all        Collection of protocols modules written in ASN.1 language (Python 3)
ii  python3-pycurl                7.45.3-1build2                    amd64      Python bindings to libcurl (Python 3)
ii  python3-pygments              2.17.2+dfsg-1                     all        syntax highlighting package written in Python 3
ii  python3-pyparsing             3.1.1-1                           all        alternative to creating and executing simple grammars - Python 3.x
ii  python3-pyrsistent:amd64      0.20.0-1build2                    amd64      persistent/functional/immutable data structures for Python
ii  python3-requests              2.31.0+dfsg-1ubuntu1.1            all        elegant and simple HTTP library for Python3, built for human beings
ii  python3-rich                  13.7.1-1                          all        render rich text, tables, progress bars, syntax highlighting, markdow
n and more
ii  python3-serial                3.5-2                             all        pyserial - module encapsulating access for the serial port
ii  python3-service-identity      24.1.0-1                          all        Service identity verification for pyOpenSSL (Python 3 module)
ii  python3-setuptools            68.1.2-2ubuntu1.2                 all        Python3 Distutils Enhancements
ii  python3-six                   1.16.0-4                          all        Python 2 and 3 compatibility library
ii  python3-software-properties   0.99.49.3                         all        manage the repositories that you install software from
ii  python3-systemd               235-1build4                       amd64      Python 3 bindings for systemd
ii  python3-twisted               24.3.0-1ubuntu0.1                 all        Event-based framework for internet applications
ii  python3-typing-extensions     4.10.0-1                          all        Backported and Experimental Type Hints for Python
ii  python3-tz                    2024.1-2                          all        Python3 version of the Olson timezone database
ii  python3-update-manager        1:24.04.12                        all        Python 3.x module for update-manager
```

```
ii  python3-yaml                  6.0.1-2build2                     amd64      YAML parser and emitter for Python3
ii  python3-zope.interface        6.1-1build1                       amd64      Interfaces for Python3
ii  python3.12                    3.12.3-1ubuntu0.11                amd64      Interactive high-level object-oriented language (version 3.12)
ii  python3.12-minimal            3.12.3-1ubuntu0.11                amd64      Minimal subset of the Python language (version 3.12)
Python: a powerful open-source programming language used everywhere
python3 is installed.
ii  libpython3-stdlib:amd64       3.12.3-0ubuntu2.1                 amd64      interactive high-level object-oriented language (default python3 vers
ion)
ii  libpython3.12-minimal:amd64   3.12.3-1ubuntu0.11                amd64      Minimal subset of the Python language (version 3.12)
ii  libpython3.12-stdlib:amd64    3.12.3-1ubuntu0.11                amd64      Interactive high-level object-oriented language (standard library, ve
rsion 3.12)
ii  libpython3.12t64:amd64        3.12.3-1ubuntu0.11                amd64      Shared Python runtime library (version 3.12)
ii  python3                       3.12.3-0ubuntu2.1                 amd64      interactive high-level object-oriented language (default python3 vers
ion)
ii  python3-apport                2.28.1-0ubuntu3.8                 all        Python 3 library for Apport crash report handling
ii  python3-apt                   2.7.7ubuntu5.2                    amd64      Python 3 interface to libapt-pkg
ii  python3-attr                  23.2.0-2                          all        Attributes without boilerplate (Python 3)
ii  python3-automat               22.10.0-2                         all        Self-service finite-state machines for the programmer on the go
ii  python3-babel                 2.10.3-3build1                    all        tools for internationalizing Python applications - Python 3.x
ii  python3-bcrypt                3.2.2-1build1                     amd64      password hashing library for Python 3
ii  python3-blinker               1.7.0-1                           all        Fast, simple object-to-object and broadcast signaling (Python3)
ii  python3-certifi               2023.11.17-1                      all        root certificates for validating SSL certs and verifying TLS hosts (p
ython3)
ii  python3-cffi-backend:amd64    1.16.0-2build1                    amd64      Foreign Function Interface for Python 3 calling C code - runtime
ii  python3-chardet               5.2.0+dfsg-1                      all        Universal Character Encoding Detector (Python3)
ii  python3-click                 8.1.6-2                           all        Wrapper around optparse for command line utilities - Python 3.x
ii  python3-colorama              0.4.6-4                           all        Cross-platform colored terminal text in Python - Python 3.x
ii  python3-commandnotfound       23.04.0                           all        Python 3 bindings for command-not-found.
ii  python3-configobj             5.0.8-3                           all        simple but powerful config file reader and writer for Python 3
ii  python3-constantly            23.10.4-1                         all        Symbolic constants in Python
ii  python3-cryptography          41.0.7-4ubuntu0.1                 amd64      Python library exposing cryptographic recipes and primitives (Python
3)
ii  python3-dbus                  1.3.2-5build3                     amd64      simple interprocess messaging system (Python 3 interface)
ii  python3-debconf               1.5.86ubuntu1                     all        interact with debconf from Python 3
ii  python3-distro                1.9.0-1                           all        Linux OS platform information API
ii  python3-distro-info           1.7build1                         all        information about distributions' releases (Python 3 module)
ii  python3-distupgrade           1:24.04.28                        all        manage release upgrades
ii  python3-gdbm:amd64            3.12.3-0ubuntu1                   amd64      GNU dbm database support for Python 3.x
ii  python3-gi                    3.48.2-1                          amd64      Python 3 bindings for gobject-introspection libraries
ii  python3-hamcrest              2.1.0-1                           all        Hamcrest framework for matcher objects (Python 3)
ii  python3-httplib2              0.20.4-3                          all        comprehensive HTTP client library written for Python3
```

```
ii  python3-incremental          22.10.0-1                       all        Library for versioning Python projects
ii  python3-jinja2               3.1.2-1ubuntu1.3                all        small but fast and easy to use stand-alone template engine
ii  python3-json-pointer         2.0-0ubuntu1                    all        resolve JSON pointers - Python 3.x
ii  python3-jsonpatch            1.32-3                          all        library to apply JSON patches - Python 3.x
ii  python3-jsonschema           4.10.3-2ubuntu1                 all        An(other) implementation of JSON Schema (Draft 3, 4, 6, 7)
ii  python3-jwt                  2.7.0-1                         all        Python 3 implementation of JSON Web Token
ii  python3-launchpadlib         1.11.0-6                        all        Launchpad web services client library (Python 3)
ii  python3-lazr.restfulclient   0.14.6-1                        all        client for lazr.restful-based web services (Python 3)
ii  python3-lazr.uri             1.0.6-3                         all        library for parsing, manipulating, and generating URIs
ii  python3-markdown-it          3.0.0-2                         all        Python port of markdown-it and some its associated plugins
ii  python3-markupsafe           2.1.5-1build2                   amd64      HTML/XHTML/XML string library
ii  python3-mdurl                0.1.2-1                         all        Python port of the JavaScript mdurl package
ii  python3-minimal              3.12.3-0ubuntu2.1               amd64      minimal subset of the Python language (default python3 version)
ii  python3-netifaces:amd64      0.11.0-2build3                  amd64      portable network interface information - Python 3.x
ii  python3-netplan              1.1.2-8ubuntu1~24.04.1          amd64      Declarative network configuration Python bindings
ii  python3-newt:amd64           0.52.24-2ubuntu2                amd64      NEWT module for Python3
ii  python3-oauthlib             3.2.2-1                         all        generic, spec-compliant implementation of OAuth for Python3
ii  python3-openssl              23.2.0-1                        all        Python 3 wrapper around the OpenSSL library
ii  python3-pkg-resources        68.1.2-2ubuntu1.2               all        Package Discovery and Resource Access using pkg_resources
ii  python3-problem-report       2.28.1-0ubuntu3.8               all        Python 3 library to handle problem reports
ii  python3-pyasn1               0.4.8-4ubuntu0.1                all        ASN.1 library for Python (Python 3 module)
ii  python3-pyasn1-modules       0.2.8-1                         all        Collection of protocols modules written in ASN.1 language (Python 3)
ii  python3-pycurl               7.45.3-1build2                  amd64      Python bindings to libcurl (Python 3)
ii  python3-pygments             2.17.2+dfsg-1                   all        syntax highlighting package written in Python 3
ii  python3-pyparsing            3.1.1-1                         all        alternative to creating and executing simple grammars - Python 3.x
ii  python3-pyrsistent:amd64     0.20.0-1build2                  amd64      persistent/functional/immutable data structures for Python
ii  python3-requests             2.31.0+dfsg-1ubuntu1.1          all        elegant and simple HTTP library for Python3, built for human beings
ii  python3-rich                 13.7.1-1                        all        render rich text, tables, progress bars, syntax highlighting, markdow
n and more
ii  python3-serial               3.5-2                           all        pyserial - module encapsulating access for the serial port
ii  python3-service-identity     24.1.0-1                        all        Service identity verification for pyOpenSSL (Python 3 module)
ii  python3-setuptools           68.1.2-2ubuntu1.2               all        Python3 Distutils Enhancements
ii  python3-six                  1.16.0-4                        all        Python 2 and 3 compatibility library
ii  python3-software-properties  0.99.49.3                       all        manage the repositories that you install software from
ii  python3-systemd              235-1build4                     amd64      Python 3 bindings for systemd
ii  python3-twisted              24.3.0-1ubuntu0.1               all        Event-based framework for internet applications
ii  python3-typing-extensions    4.10.0-1                        all        Backported and Experimental Type Hints for Python
ii  python3-tz                   2024.1-2                        all        Python3 version of the Olson timezone database
ii  python3-update-manager       1:24.04.12                      all        Python 3.x module for update-manager
ii  python3-urllib3              2.0.7-1ubuntu0.6                all        HTTP library with thread-safe connection pooling for Python3
ii  python3-wadllib              1.3.6-5                         all        Python 3 library for navigating WADL files
```

```
loq@SARANG:~$ nano script2.sh
loq@SARANG:~$ chmod +x script2.sh
loq@SARANG:~$ ./script2.sh
python3 is installed.
ii  libpython3-stdlib:amd64      3.12.3-0ubuntu2.1               amd64      interactive high-level object-oriented language (default python3 vers
ion)
ii  libpython3.12-minimal:amd64  3.12.3-1ubuntu0.11               amd64      Minimal subset of the Python language (version 3.12)
ii  libpython3.12-stdlib:amd64   3.12.3-1ubuntu0.11               amd64      Interactive high-level object-oriented language (standard library, ve
rsion 3.12)
ii  libpython3.12t64:amd64       3.12.3-1ubuntu0.11               amd64      Shared Python runtime library (version 3.12)
ii  python3                      3.12.3-0ubuntu2.1                amd64      interactive high-level object-oriented language (default python3 vers
ion)
ii  python3-apport               2.28.1-0ubuntu3.8                all        Python 3 library for Apport crash report handling
ii  python3-apt                  2.7.7ubuntu5.2                   amd64      Python 3 interface to libapt-pkg
ii  python3-attr                 23.2.0-2                         all        Attributes without boilerplate (Python 3)
ii  python3-automat              22.10.0-2                        all        Self-service finite-state machines for the programmer on the go
ii  python3-babel                2.10.3-3build1                   all        tools for internationalizing Python applications - Python 3.x
ii  python3-bcrypt               3.2.2-1build1                    amd64      password hashing library for Python 3
ii  python3-blinker              1.7.0-1                          all        Fast, simple object-to-object and broadcast signaling (Python3)
ii  python3-certifi              2023.11.17-1                     all        root certificates for validating SSL certs and verifying TLS hosts (p
ython3)
ii  python3-cffi-backend:amd64   1.16.0-2build1                   amd64      Foreign Function Interface for Python 3 calling C code - runtime
ii  python3-chardet              5.2.0+dfsg-1                     all        Universal Character Encoding Detector (Python3)
ii  python3-click                8.1.6-2                          all        Wrapper around optparse for command line utilities - Python 3.x
ii  python3-colorama             0.4.6-4                          all        Cross-platform colored terminal text in Python - Python 3.x
ii  python3-commandnotfound      23.04.0                          all        Python 3 bindings for command-not-found.
ii  python3-configobj            5.0.8-3                          all        simple but powerful config file reader and writer for Python 3
ii  python3-constantly           23.10.4-1                        all        Symbolic constants in Python
ii  python3-cryptography         41.0.7-4ubuntu0.1                amd64      Python library exposing cryptographic recipes and primitives (Python
3)
ii  python3-dbus                 1.3.2-5build3                    amd64      simple interprocess messaging system (Python 3 interface)
ii  python3-debconf              1.5.86ubuntu1                    all        interact with debconf from Python 3
ii  python3-distro               1.9.0-1                          all        Linux OS platform information API
ii  python3-distro-info          1.7build1                        all        information about distributions' releases (Python 3 module)
ii  python3-distupgrade          1:24.04.28                       all        manage release upgrades
ii  python3-gdbm:amd64           3.12.3-0ubuntu1                  amd64      GNU dbm database support for Python 3.x
ii  python3-gi                   3.48.2-1                         amd64      Python 3 bindings for gobject-introspection libraries
ii  python3-hamcrest             2.1.0-1                          all        Hamcrest framework for matcher objects (Python 3)
ii  python3-httplib2             0.20.4-3                         all        comprehensive HTTP client library written for Python3
ii  python3-hyperlink            21.0.0-5                         all        Immutable, Pythonic, correct URLs.
ii  python3-idna                 3.6-2ubuntu0.1                   all        Python IDNA2008 (RFC 5891) handling (Python 3)
```

This script checks whether python3 is installed in the system and displays package details. It uses conditional statements and a case statement.

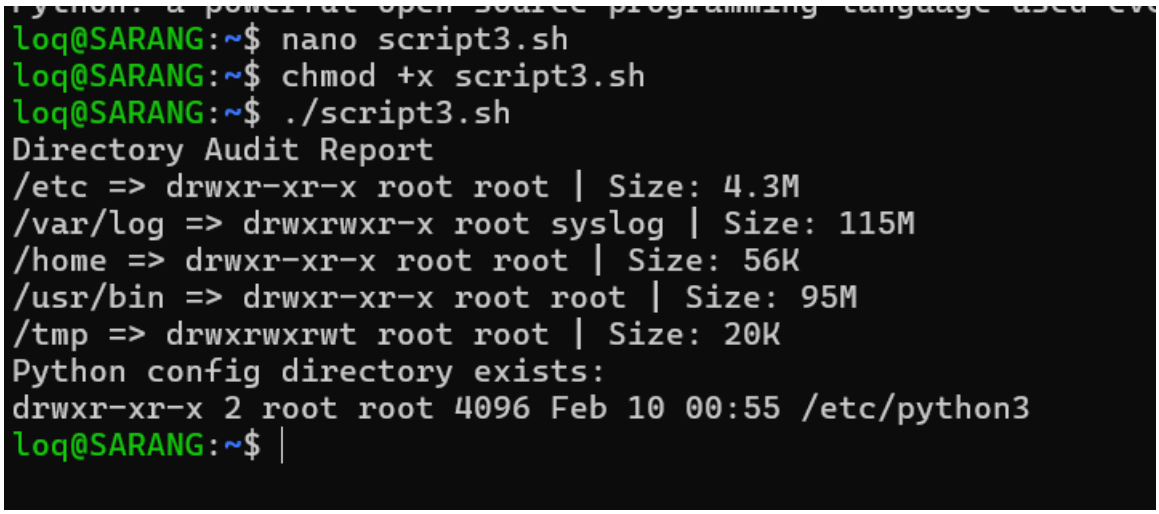
Script 3 – Disk and Permission Auditor

Author: Sarang Kulkarni

#!/bin/bash

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

```
echo "Directory Audit Report"


for DIR in "${DIRS[@]}"; do
  if [ -d "$DIR" ]; then
    PERMS=$(ls -ld $DIR | awk '{print $1, $3, $4}')
    SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)
    echo "$DIR => $PERMS | Size: $SIZE"
  fi
done


# Python config check
if [ -d "/etc/python3" ]; then
  echo "Python config directory exists:"
  ls -ld /etc/python3
fi
```

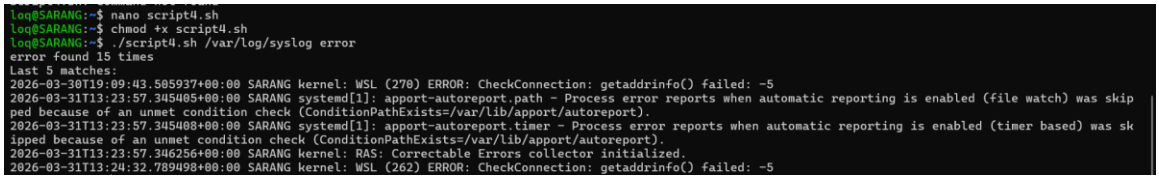


This script checks important directories and displays permissions and disk usage using a loop.

Script 4 – Log File Analyzer

Author: Sarang Kulkarni

```
#!/bin/bash


LOGFILE=$1

KEYWORD=${2:-"error"}

COUNT=0


if [ ! -f "$LOGFILE" ]; then

   echo "File not found"

   exit 1

fi


while IFS= read -r LINE; do

   if echo "$LINE" | grep -iq "$KEYWORD"; then

      COUNT=$((COUNT + 1))

   fi

done < "$LOGFILE"


echo "$KEYWORD found $COUNT times"


echo "Last 5 matches:"

grep -i "$KEYWORD" "$LOGFILE" | tail -5
```

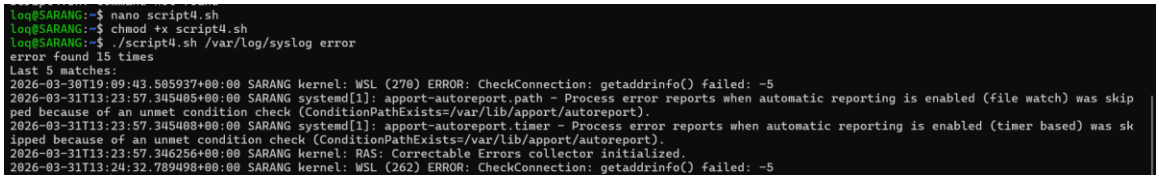


This script reads a log file and counts how many times a keyword appears.

Script 5 – Open Source Manifesto Generator

Author: Sarang Kulkarni

```
#!/bin/bash


read -p "Tool: " TOOL

read -p "Freedom means: " FREEDOM

read -p "What will you build: " BUILD


DATE=$(date)

OUTPUT="manifesto.txt"


echo "On $DATE, I believe open source means $FREEDOM. I use $TOOL daily and I want to build $BUILD and share it with everyone." > $OUTPUT


cat $OUTPUT
```



This script takes user input and generates a personal open-source manifesto and saves it to a file.

4. Conclusion

This project helped me understand open-source software and how important it is for computing. I learned that Python is an open-source programming language that is easy to use and widely used. The project also helped me understand how Linux manages software and how shell scripting can be used to automate tasks.

Overall, this project helped me learn about open-source software, Linux systems and shell scripting. Open-source software is very important for learning, innovation and collaboration.